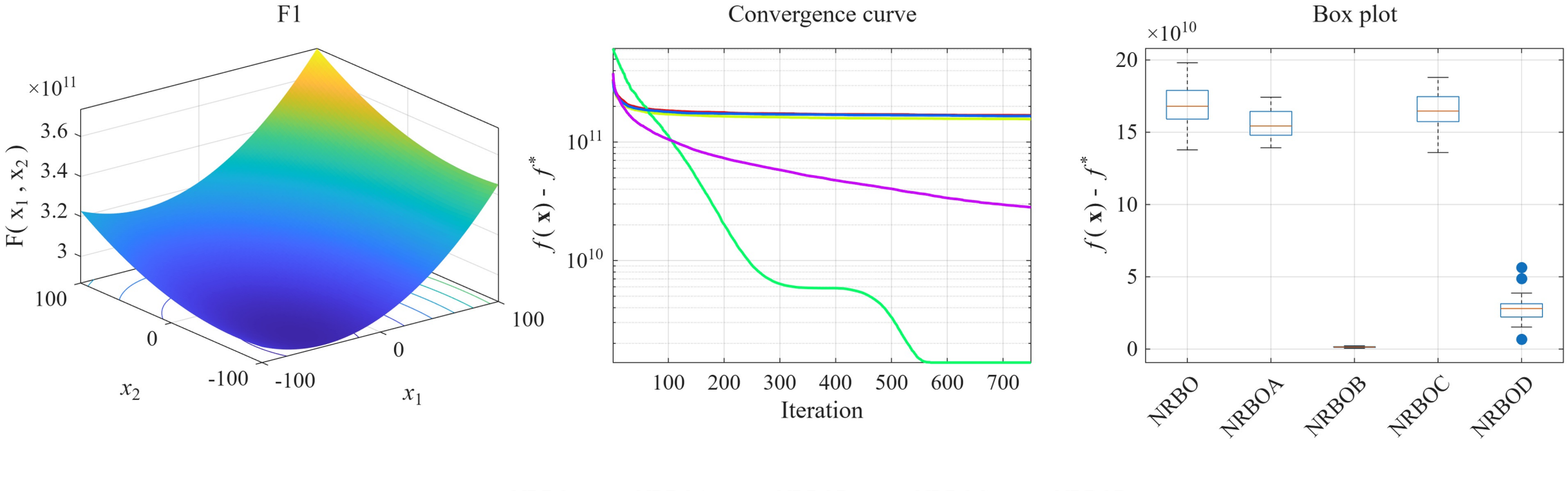

F1
×10^11
3.6
3.4
3.2
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^11
10^10
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^10
20
15
10
5
0
f(x) - f*
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD